AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 28 · WORKBOOK

Privacy by Design

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected privacy and lifecycle record.

Explain the stores, trace the threshold, audit policy scope and predict a copy failure.

Keep one habit: Keep purpose, permitted use, exact boundary, minimal audit and unresolved expiry together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Red Team the Product

Quality Rubrics

Requests Responses and JSON

W1 · Explain retention and deletion

> Name purpose, store locations, threshold and evidence.

Your evidence record

- Policies are separate: main picture delete age >= 3 days, A age 2 kept, B age 3 and C age 4 deleted plus cached B; W2 delete age >= 7 days in primary/cache, keep 6/remove 7 and 9; allowed content-free audit expires age >= 30 days, backup policy unspecified. Independent policy states 5 days. Do not retain expired content in proof logs, invent deletion success or treat these toy thresholds as legal requirements.

- Reference for this exercise: open this week's student lesson, "Privacy by Design", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the seven-day policy

> Keep age 6 and remove content ages 7/9 from primary and cache.

Your evidence record

- Policies are separate: main picture delete age >= 3 days, A age 2 kept, B age 3 and C age 4 deleted plus cached B; W2 delete age >= 7 days in primary/cache, keep 6/remove 7 and 9; allowed content-free audit expires age >= 30 days, backup policy unspecified. Independent policy states 5 days. Do not retain expired content in proof logs, invent deletion success or treat these toy thresholds as legal requirements.

- Record ages are6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit policy-scope claims

> The classroom policy is not a universal legal retention rule.

Your evidence record

- Policies are separate: main picture delete age >= 3 days, A age 2 kept, B age 3 and C age 4 deleted plus cached B; W2 delete age >= 7 days in primary/cache, keep 6/remove 7 and 9; allowed content-free audit expires age >= 30 days, backup policy unspecified. Independent policy states 5 days. Do not retain expired content in proof logs, invent deletion success or treat these toy thresholds as legal requirements.

- A report omits this qualification: This fictional **retention** exercise is not a statement of legal **retention** requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing copy failure

> Predict a residual-store outcome before the paper trace.

Your evidence record

- Policies are separate: main picture delete age >= 3 days, A age 2 kept, B age 3 and C age 4 deleted plus cached B; W2 delete age >= 7 days in primary/cache, keep 6/remove 7 and 9; allowed content-free audit expires age >= 30 days, backup policy unspecified. Independent policy states 5 days. Do not retain expired content in proof logs, invent deletion success or treat these toy thresholds as legal requirements.

- Reference for this exercise: open this week’s student lesson, “Privacy by Design”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Trace the fresh five-day lifecycle independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Apply the stated five-day policy inclusively: deletion is due when age >= 5 days. Logs are a declared store too. For this toy contract, full-content logs are not permitted audit metadata; identify required handling and residual copies rather than silently keeping them. No backup rule or actual deletion result is supplied.

Retention is5 days, inclusive **deletion boundary**. Ages4, 5, 6; a log stores complete content. Trace required action.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Create a minimal data map and visible memory controls before expanding features.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Data **retention** and deletion. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Data **retention** and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A study tool needs topic and reading level but proposes storing addresses and all conversations forever. Design a smaller record and a deletion rule.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A study tool plans to retain names, addresses and all chats forever.

Design a smaller data plan.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A practice quiz needs a topic and difficulty. A form also asks for home address and date of birth. What should be removed and why?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 28 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A study tool needs topic and reading level but proposes storing addresses and all conversations forever. Design a smaller record and a deletion rule.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Retention is5 days, inclusive **deletion boundary**. Ages4, 5, 6; a log stores complete content. Trace required action.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

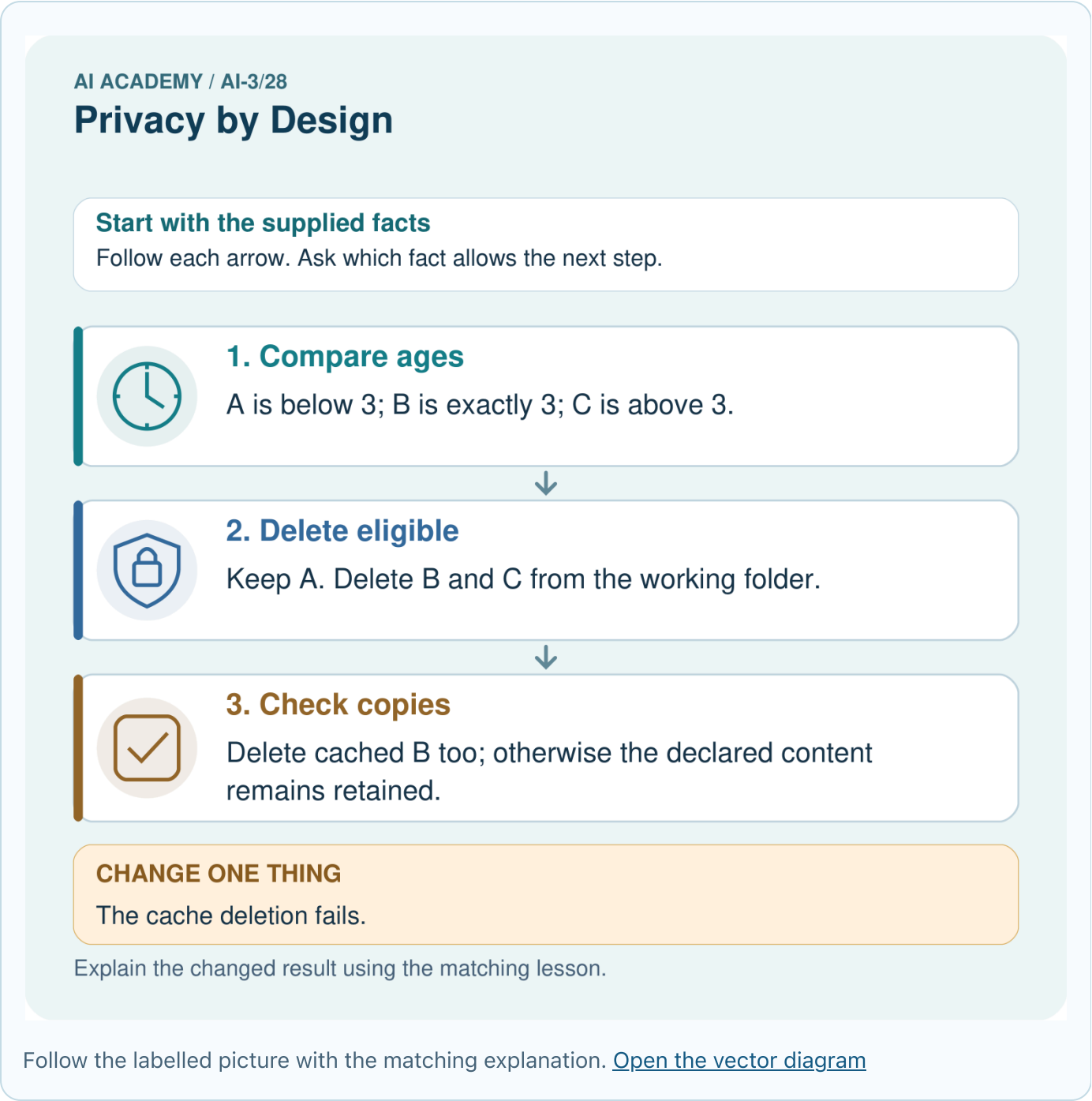
> Keep the explanation and the support you used.

Use this when helpful

Trace deletion through both a working folder and a cache.

Fictional **retention** policy: delete records aged at least 3 days. Ages A=2, B=3, C=4; the cache contains a copy of B.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which IDs and locations must appear in the deletion **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Trace deletion through both a working folder and a cache.
- Which labelled step matches this detail? Delete cached B too; otherwise the declared content remains retained.
- 1. Compare ages

- 2. Delete eligible
- 3. Check copies
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Check copies Delete cached B too; otherwise the declared content remains retained. Read the picture in order.

  Delete cached B too; otherwise the declared content remains retained. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: The cache deletion fails. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Report incomplete deletion rather than success.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: Report incomplete deletion rather than success. Removing the working copy does not remove the known retained cache copy.

> **My selected conclusion, deciding fact and explanation**
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> | |
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> | |
> | |
> | |

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Red Team the Product (ai-3/27). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Quality Rubrics (ai-3/25). Goal: Judge an output using separate observable rubric criteria. Short practice: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records. Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
> |---|---|
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> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

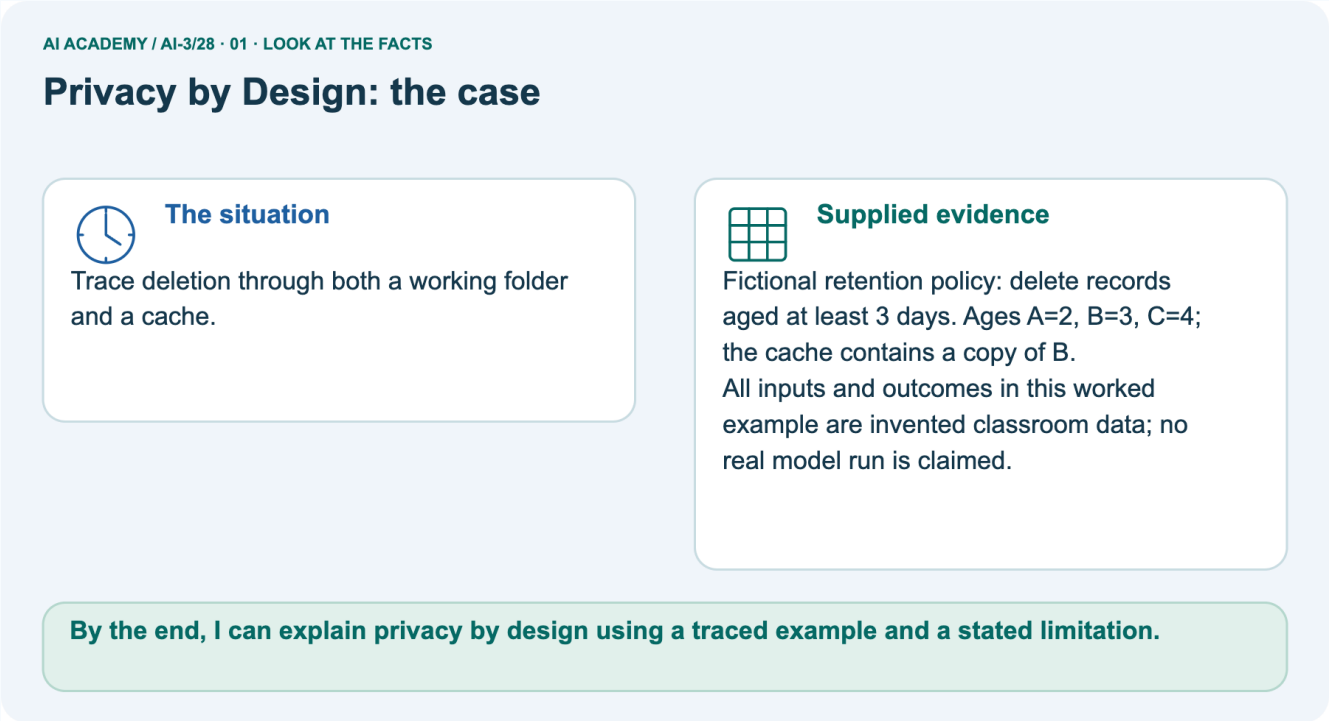
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Privacy by Design: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

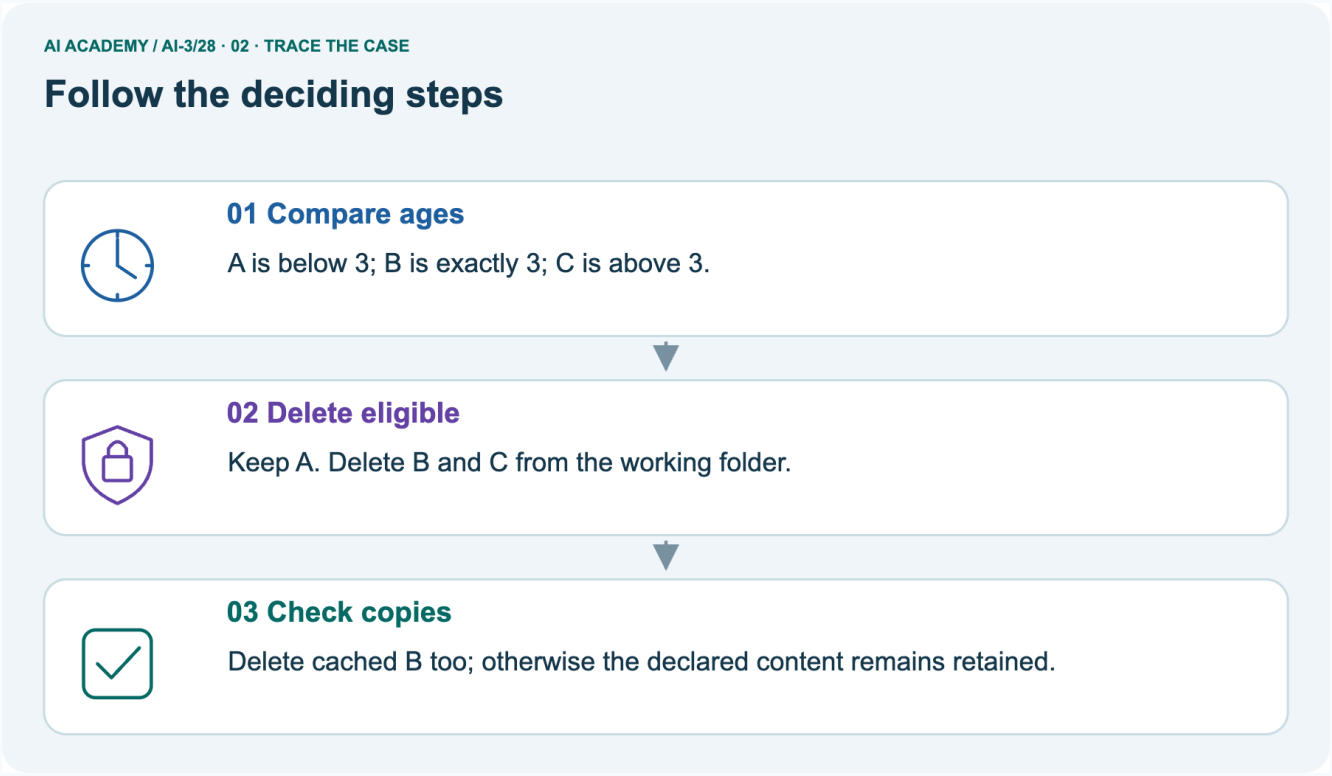
Trace deletion through both a working folder and a cache. Supplied facts: Fictional **retention** policy: delete records aged at least 3 days. Ages A=2, B=3, C=4; the cache contains a copy of B.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



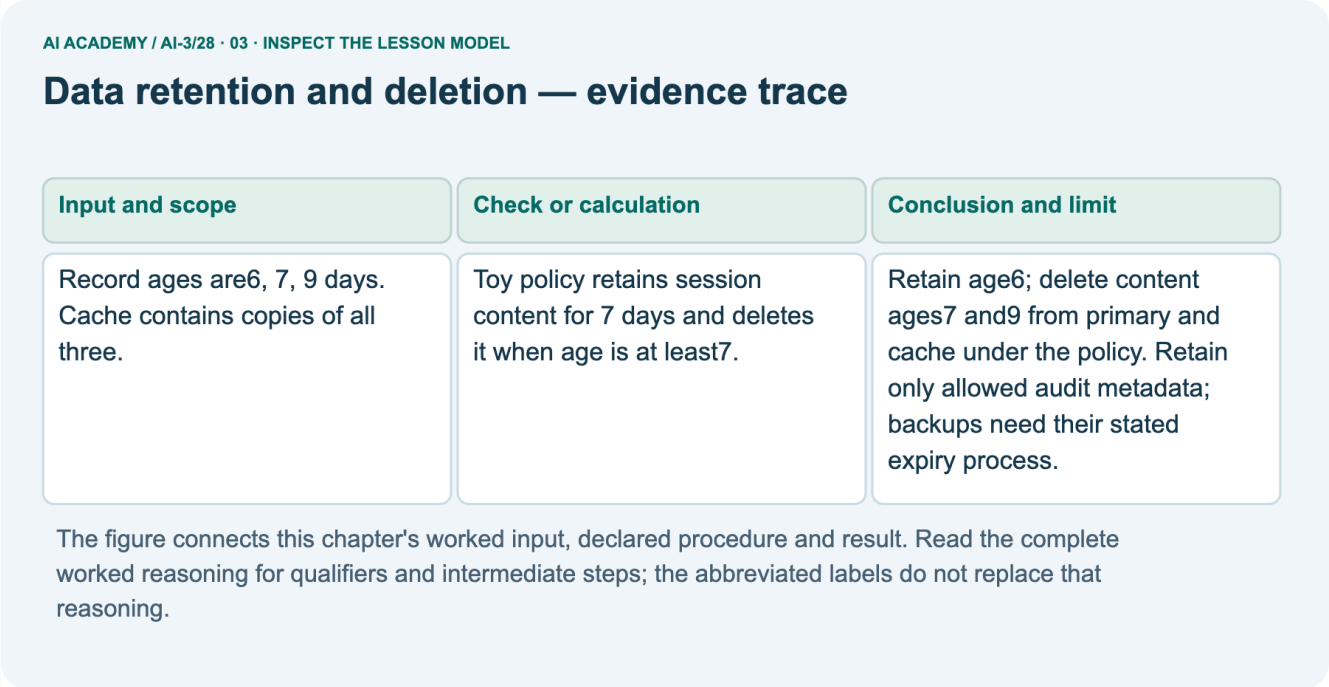
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare ages: A is below 3; B is exactly 3; C is above 3. → Delete eligible: Keep A. Delete B and C from the working folder. → Check copies: Delete cached B too; otherwise the declared content remains retained.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Data retention and deletion — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Data **retention** and deletion — **evidence** trace. Record ages are6, 7, 9 days.

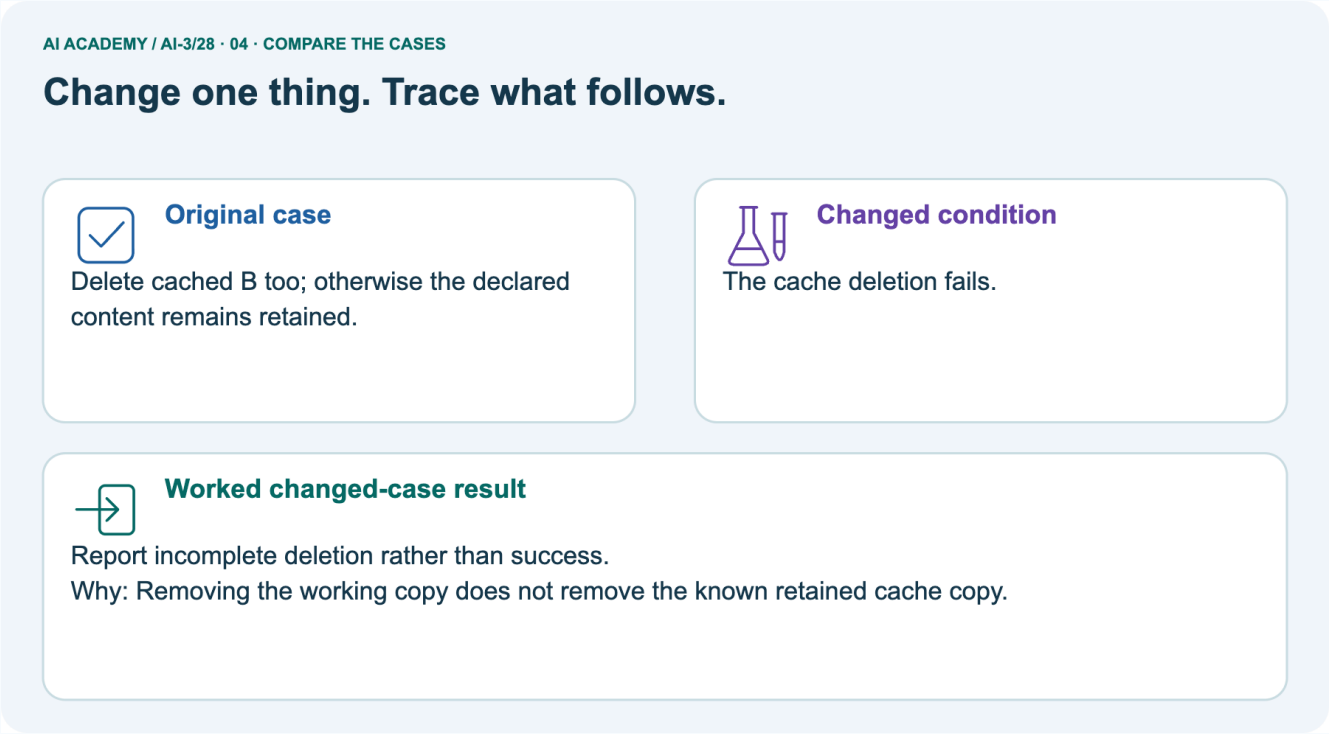
Cache contains copies of all three. | Toy policy retains session content for 7 days and deletes it when age is at least7. | Retain age6; delete content ages7 and9 from primary and cache under the policy.

Retain only allowed audit metadata; backups need their stated expiry **process**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

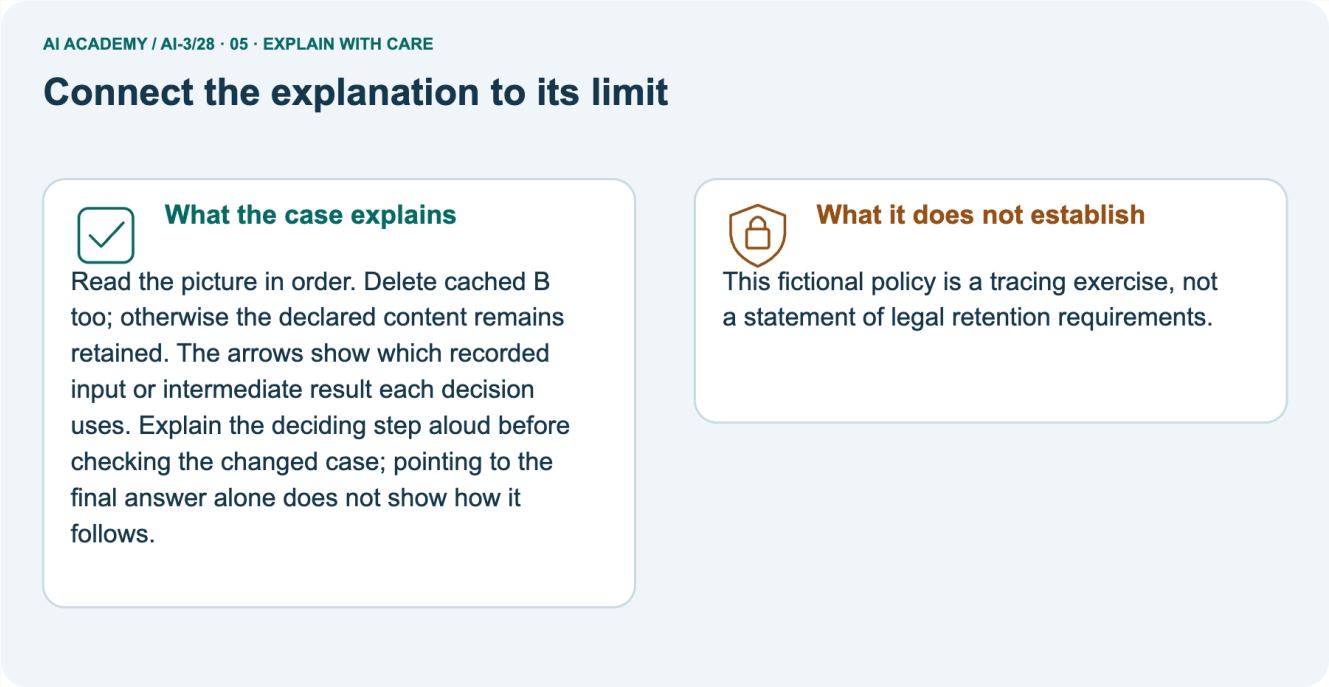
Original case: Delete cached B too; otherwise the declared content remains retained. Changed condition: The cache deletion fails. Expected result: Report incomplete deletion rather than success.

Reason: Removing the working copy does not remove the known retained cache copy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Delete cached B too; otherwise the declared content remains retained. The arrows show which recorded **input** or intermediate result each decision uses.

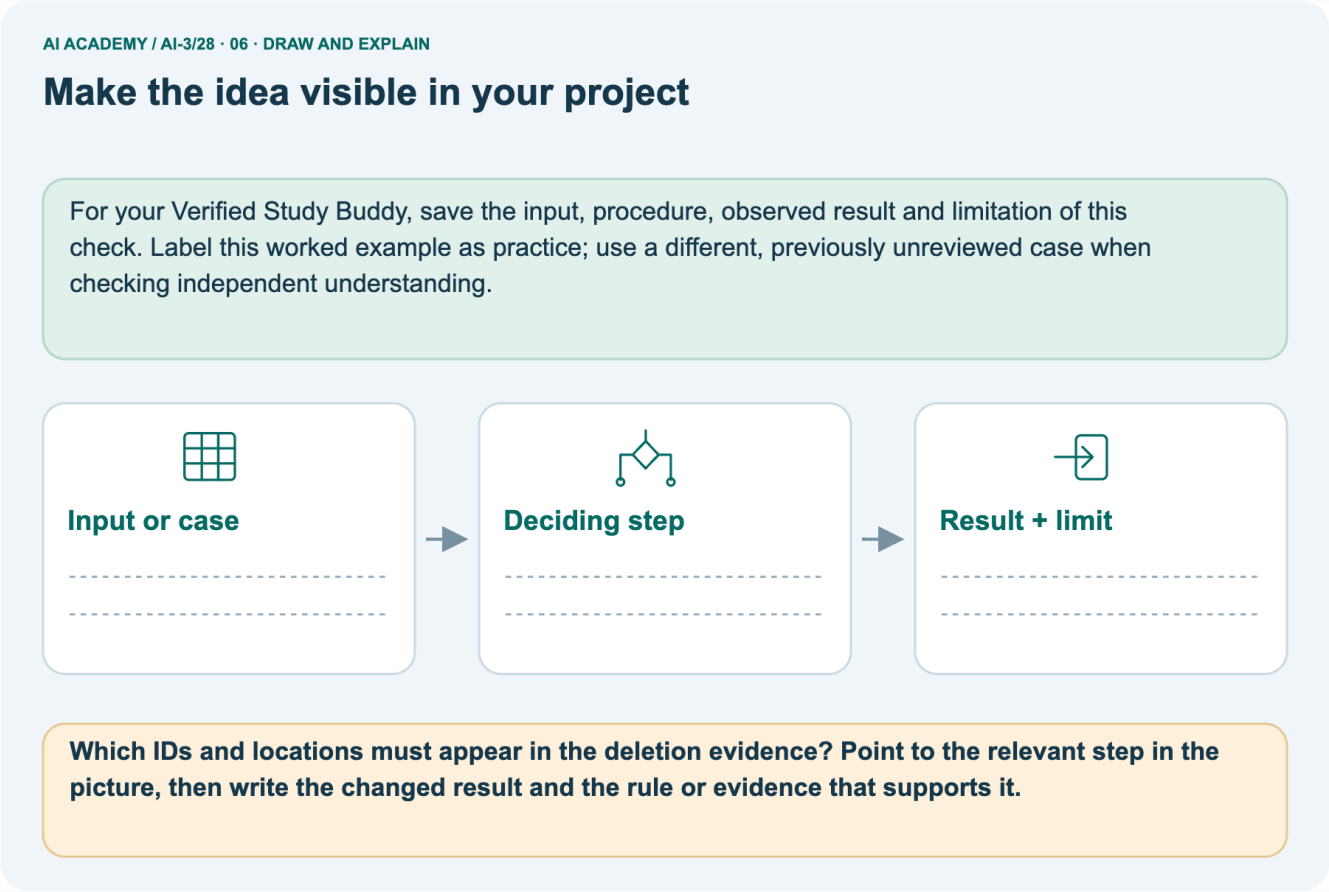
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This fictional policy is a tracing exercise, not a statement of legal **retention** requirements.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which IDs and locations must appear in the deletion **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Privacy by Design: the case

02. Follow the deciding steps

03. Data retention and deletion — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.